

# The Gunsmith's Book of Work

*Concerning the assembly of the flintlock pistol,  
the long gun, and their ammunition*



Complete Flintlock Mechanism

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Bound Pistol Stock

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Reinforced Long Barrel

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With twenty workshop plates,  
particular patterns, and a ledger of materials.

A period-inspired handbook for Minecraft 1.21.1  
TaCZ · Create · TaCZ: Creatified · KubeJS

# Of order at the bench

Lay out the several parts before beginning the work. Keep the barrel-maker's pieces apart from the stock-maker's, and gather the small fittings in a tray; much confusion begins with a misplaced screw.

This is a fictional workshop handbook for the installed Minecraft recipes. Its voice recalls an old trade manual; its materials, patterns and timings belong to the game, rather than a reconstruction of historical manufacture.

## How the plates are read

Every grid is seen from the front. A letter names one item in that square. An empty square must remain empty. Letter meanings change from plate to plate: consult each plate's key.

Every component recipe gives **one item**. A Gun Screws icon depicts several screws, but each craft gives only one inventory item. Screws, precision mechanisms and ramrods are consumed as ingredients.

## The several stations

- **Crafting table:** all shaped components and both final guns.
- **Blast furnace:** tempered steel and the tempered mainspring.
- **Mechanical Crafters:** final ammunition, in a powered 3×3 array.

The twenty recipes are intentionally absent from EMI/JEI, even for operators. Possession of this handbook supplies their patterns; the actual recipes remain craftable.

# Before the work begins

These quantities reckon every custom part down to ordinary supplies and finished Create materials. Stock the bench accordingly.

| FINISHED SUPPLY       | PISTOL | RIFLE | BOTH |
|-----------------------|--------|-------|------|
| Iron Sheet            | 76     | 125   | 201  |
| Charcoal              | 56     | 96    | 152  |
| Andesite Alloy        | 16     | 28    | 44   |
| Iron Nugget           | 26     | 40    | 66   |
| Precision Mechanism   | 3      | 6     | 9    |
| Flint                 | 2      | 2     | 4    |
| Brass Sheet           | 9      | 24    | 33   |
| Tripwire Hook         | 1      | 1     | 2    |
| Stripped Dark Oak Log | 6      | 18    | 24   |
| Leather               | 3      | 5     | 8    |

Sheets, alloys and precision mechanisms are counted as finished supplies. Their upstream ingredients, machine construction, furnace fuel and ammunition are additional. Do not count intermediate components again as extra costs.

## Preparing Create materials

Press iron and brass ingots into their respective sheets. Heated Basin mixing combines one copper and one zinc ingot into two brass ingots. The ordinary alloy crafting pattern uses two andesite and two iron or zinc nuggets.

For a precision mechanism, begin with a Golden Sheet. Deploy a Cogwheel, a Large Cogwheel and an Iron Nugget, in that order, five times. An attempt uses five of each applied ingredient and may yield a byproduct. Prepare enough successful mechanisms before assembly.

# Count the smaller work

The tally includes parts later consumed within other assemblies. Follow the plates to make these quantities; they are already included in the stores ledger.

| COMPONENT                    | PISTOL | RIFLE |
|------------------------------|--------|-------|
| Gun Steel Blank              | 14     | 24    |
| Tempered Gun Steel           | 14     | 24    |
| Gun Screws                   | 13     | 20    |
| Brass Barrel Band            | 1      | 6     |
| Steel Ramrod                 | 1      | 1     |
| Barrel Blank                 | 1      | 2     |
| Finished Pistol Barrel       | 1      | 2     |
| Reinforced Long Barrel       | 0      | 1     |
| Mainspring Blank             | 1      | 1     |
| Tempered Mainspring          | 1      | 1     |
| Flintlock Hammer             | 1      | 1     |
| Brass Flash Pan              | 1      | 1     |
| Trigger Assembly             | 1      | 1     |
| Complete Flintlock Mechanism | 1      | 1     |
| Hardwood Stock Blank         | 1      | 3     |
| Bound Pistol Stock           | 1      | 1     |
| Reinforced Rifle Stock       | 0      | 1     |

# The short arm



Finished Pistol Barrel



Complete Flintlock Mechanism



Bound Pistol Stock

1. Prepare 14 tempered steel pieces and 13 Gun Screws.
2. Prepare one tempered mainspring, then the hammer, flash pan and trigger. Bring these together into one complete lock.
3. Make a barrel blank and finish it into one pistol barrel.
4. Carve one hardwood blank and bind it into the pistol stock.
5. Prepare one brass band and one ramrod.
6. Join the assemblies on Plate XVIII, reserving the extra screw, tempered steel and leather required there.

The pistol consumes three precision mechanisms: one in the barrel, one in the trigger, and one in the lock.

The finished weapon is the FK15P, gun ID qk1:fk15p. It is produced unloaded.

# The shoulder arm



Reinforced Long Barrel



Complete Flintlock Mechanism



Reinforced Rifle Stock

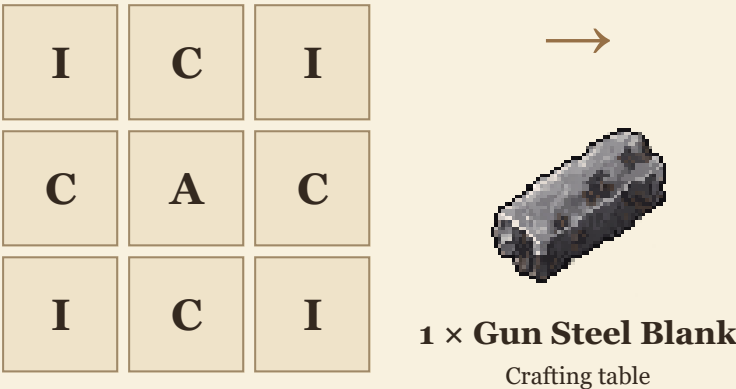
1. Prepare 24 tempered steel pieces, 20 Gun Screws and one tempered mainspring.
2. Assemble one complete lock through the hammer, pan and trigger plates.
3. Finish two pistol barrels, then join them into the reinforced long barrel.
4. Make three hardwood blanks. Bind one into a pistol stock; combine it with the other two for the rifle stock.
5. Prepare six brass bands in total: two for the long barrel, two for the rifle stock, and two for final assembly.
6. Prepare a ramrod, then complete the rifle using its final plate.

Six precision mechanisms are consumed throughout this chain. A completed pistol is never an ingredient: the rifle shares its component patterns.

The finished weapon is the FK15, gun ID qk1:fk15. It is produced unloaded.

# Gun Steel Blank

Begin with the stock of the forge. Lay iron sheets about charcoal and an andesite-alloy heart, following the nine-square pattern. The rough billet is but a beginning; set it aside for the blast furnace before using it in the finer work.

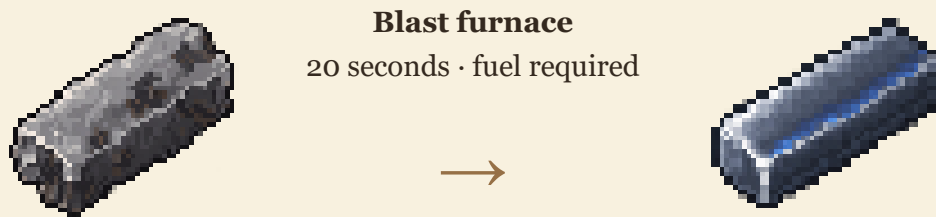


| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| I    | Iron Sheet         | 4     |
| C    | Charcoal           | 4     |
| A    | Andesite Alloy     | 1     |

Workshop record: kubejs:flintlocks/gun\_steel\_blank

# Tempered Gun Steel

Bring the rough billet to the furnace and allow the work its full appointed time. In this workshop, the blast furnace yields the tempered piece required by barrels, hammers and fittings. Count the finished pieces before beginning assembly.



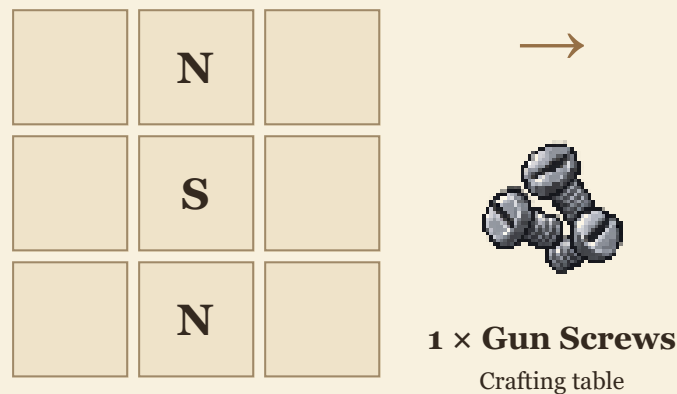
Supply one Gun Steel Blank. The recipe returns one Tempered Gun Steel. Its duration is 400 game ticks at normal server speed.

Use a blast furnace. An ordinary furnace does not perform this custom recipe.

Workshop record: kubejs:flintlocks/tempered\_gun\_steel

# Gun Screws

Small fastenings govern the larger work. Place a sheet between two iron nuggets, one above and one below. Make a reserve at the outset: many later assemblies call for these screws, and each craft yields one inventory item.



| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| N    | Iron Nugget        | 2     |
| S    | Iron Sheet         | 1     |

Workshop record: kubejs:flintlocks/gun\_screws

# Brass Barrel Band

The brass band binds the larger assemblies together. Arrange three brass sheets about the open centre and place a screw below. The rifle consumes six bands throughout its construction; the pistol requires one.



1 × Brass Barrel  
Band

Crafting table

| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| B    | Brass Sheet        | 3     |
| S    | Gun Screws         | 1     |

Workshop record: kubejs:flintlocks/barrel\_band

# Steel Ramrod

Set two tempered pieces along the rising diagonal, with a screw at its upper end. Keep the completed rod beside the final assembly: the rod is consumed as a fitted part of the finished gun.



| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| S    | Gun Screws         | 1     |
| I    | Tempered Gun Steel | 2     |

Workshop record: kubejs:flintlocks/ramrod

# Barrel Blank

Now attend to the barrel stock. Six tempered pieces form the sides of the pattern, while two iron sheets occupy its ends. Leave the middle square empty. This rough part must still pass through the finishing recipe.



| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| S    | Tempered Gun Steel | 6     |
| I    | Iron Sheet         | 2     |

Workshop record: kubejs:flintlocks/barrel\_blank

# Finished Pistol Barrel

Finish the short barrel by joining the blank with two pieces of andesite alloy, a screw and a precision mechanism. For the long gun, prepare two such barrels before proceeding to the next plate.

**B**  


**A**

**S**  


**A**

**C**

→



**1 × Finished Pistol Barrel**  
Crafting table

| MARK     | LAY UPON THE BENCH  | COUNT |
|----------|---------------------|-------|
| <b>B</b> | Barrel Blank        | 1     |
| <b>A</b> | Andesite Alloy      | 2     |
| <b>S</b> | Gun Screws          | 1     |
| <b>C</b> | Precision Mechanism | 1     |

Workshop record: kubejs:flintlocks/pistol\_barrel

# Reinforced Long Barrel

Lay the two finished short barrels at opposite ends of the grid. Tempered steel braces the corners, brass bands flank the centre, and a precision mechanism occupies the heart. Together these make the reinforced long barrel.

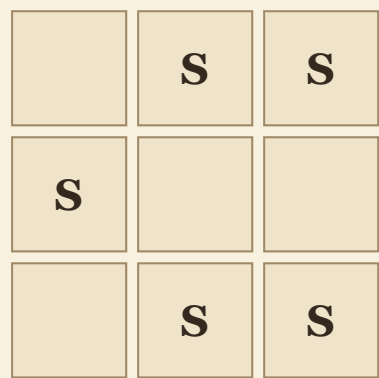


| MARK | LAY UPON THE BENCH     | COUNT |
|------|------------------------|-------|
| S    | Tempered Gun Steel     | 4     |
| B    | Finished Pistol Barrel | 2     |
| I    | Brass Barrel Band      | 2     |
| P    | Precision Mechanism    | 1     |

Workshop record: kubejs:flintlocks/rifle\_barrel

# Mainspring Blank

A lock begins with its spring. Arrange five iron sheets in the open, bent pattern shown on this plate. Do not fill the empty squares. The resulting blank still requires the furnace before it belongs in the trigger assembly.

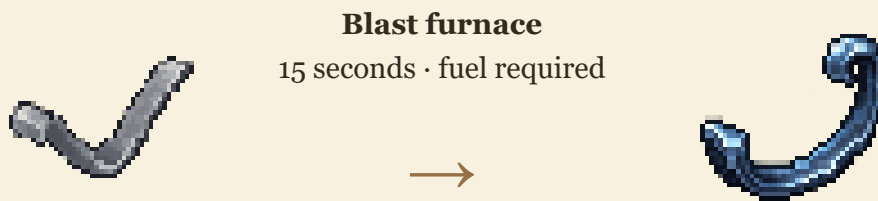


| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| S    | Iron Sheet         | 5     |

Workshop record: kubejs:flintlocks/mainspring\_blank

# Tempered Mainspring

Give the spring blank its turn in the blast furnace. The finished spring is the dark, curved piece shown here. One serves the trigger assembly of either firearm; reserve it for that work.



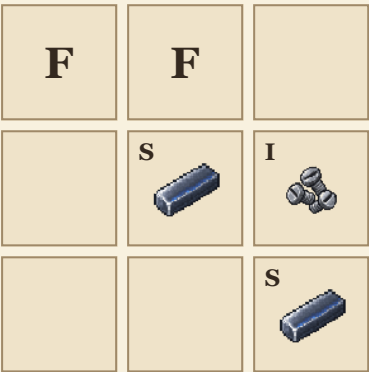
Supply one Mainspring Blank. The recipe returns one Tempered Mainspring. Its duration is 300 game ticks at normal server speed.

Use a blast furnace. An ordinary furnace does not perform this custom recipe.

Workshop record: kubejs:flintlocks/tempered\_mainspring

# Flintlock Hammer

The hammer is the raised member of the lock. Two flints, two tempered pieces and a screw are laid in the stepped pattern. Assemble it as its own component before bringing it to the lock plate.



| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| F    | Flint              | 2     |
| S    | Tempered Gun Steel | 2     |
| I    | Gun Screws         | 1     |

Workshop record: kubejs:flintlocks/flintlock\_hammer

# Brass Flash Pan

Prepare the brass pan from four brass sheets, a tempered piece and a screw. Its golden bowl distinguishes it from the hammer and trigger. Store these small lock parts together to avoid confusing the later assembly.

B

B

B

S  


B

I  


→



1 × Brass Flash Pan

Crafting table

| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| B    | Brass Sheet        | 4     |
| S    | Tempered Gun Steel | 1     |
| I    | Gun Screws         | 1     |

Workshop record: kubejs:flintlocks/flash\_pan

# Trigger Assembly

Set the tempered spring above the precision mechanism, flank the mechanism with screws, and place the tripwire hook below. The hook is a Minecraft ingredient; this pattern is the workshop's game recipe.

S

I

P

I

H

→



1 × Trigger Assembly

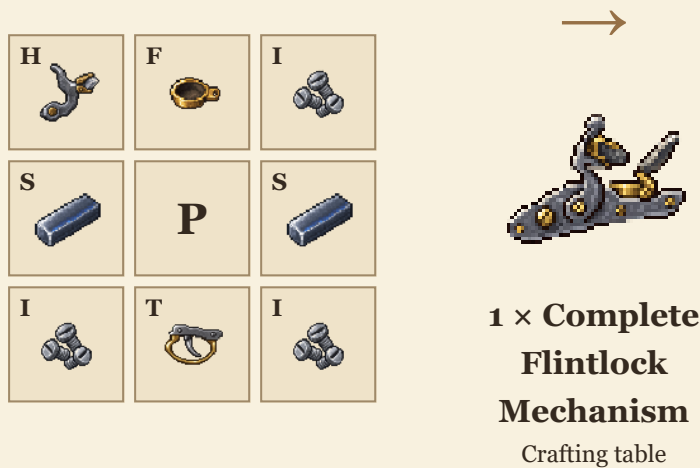
Crafting table

| MARK | LAY UPON THE BENCH  | COUNT |
|------|---------------------|-------|
| S    | Tempered Mainspring | 1     |
| I    | Gun Screws          | 2     |
| P    | Precision Mechanism | 1     |
| H    | Tripwire Hook       | 1     |

Workshop record: kubejs:flintlocks/trigger\_assembly

# Complete Flintlock Mechanism

Gather hammer, pan and trigger before the lock is assembled. Lay them around the central precision mechanism with tempered steel and screws. The completed lock is common to both pistol and rifle.

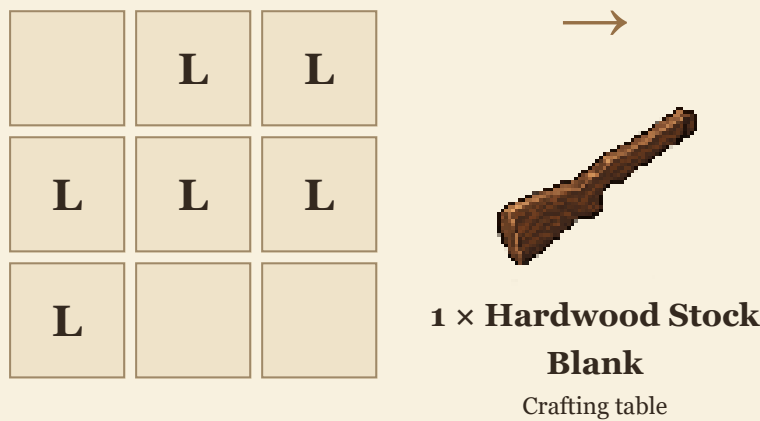


| MARK | LAY UPON THE BENCH  | COUNT |
|------|---------------------|-------|
| H    | Flintlock Hammer    | 1     |
| F    | Brass Flash Pan     | 1     |
| I    | Gun Screws          | 3     |
| S    | Tempered Gun Steel  | 2     |
| P    | Precision Mechanism | 1     |
| T    | Trigger Assembly    | 1     |

Workshop record: kubejs:flintlocks/flintlock\_mechanism

# Hardwood Stock Blank

Turn next to the stock-maker's bench. Six stripped dark oak logs form the stepped blank. Use the exact timber named in the ledger: planks, unstripped logs and other species do not answer this pattern.

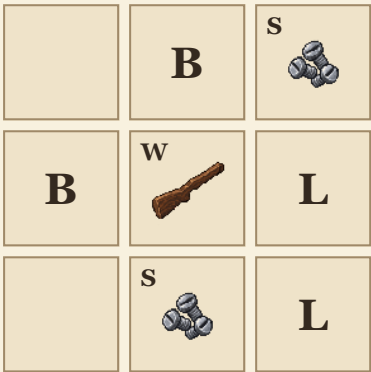


| MARK | LAY UPON THE BENCH    | COUNT |
|------|-----------------------|-------|
| L    | Stripped Dark Oak Log | 6     |

Workshop record: kubejs:flintlocks/stock\_blank

# Bound Pistol Stock

Bind the short stock with leather and furnish it with brass and screws. Its curved grip marks it as the pistol stock. The rifle also uses one such stock as an intermediate, so do not mistake it for a completed pistol.

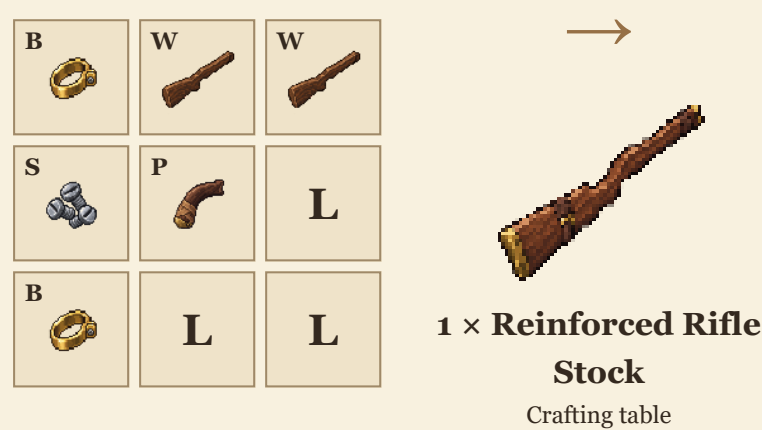


| MARK | LAY UPON THE BENCH   | COUNT |
|------|----------------------|-------|
| B    | Brass Sheet          | 2     |
| S    | Gun Screws           | 2     |
| W    | Hardwood Stock Blank | 1     |
| L    | Leather              | 2     |

Workshop record: kubejs:flintlocks/pistol\_stock

# Reinforced Rifle Stock

For the shoulder arm, gather two further hardwood blanks and the bound pistol stock. Add leather, a screw and two brass bands. The resulting reinforced stock is reserved for the final long-gun assembly.



| MARK | LAY UPON THE BENCH   | COUNT |
|------|----------------------|-------|
| B    | Brass Barrel Band    | 2     |
| W    | Hardwood Stock Blank | 2     |
| S    | Gun Screws           | 1     |
| P    | Bound Pistol Stock   | 1     |
| L    | Leather              | 3     |

Workshop record: kubejs:flintlocks/rifle\_stock

# FK15P Flintlock Pistol

Bring the barrel, complete lock, ramrod and bound stock to the bench. Place the band, screw, tempered piece and leather precisely as shown. The work yields one FK15P pistol, unloaded and ready for its Minecraft ammunition.



| MARK | LAY UPON THE BENCH           | COUNT |
|------|------------------------------|-------|
| B    | Finished Pistol Barrel       | 1     |
| M    | Complete Flintlock Mechanism | 1     |
| R    | Steel Ramrod                 | 1     |
| S    | Gun Screws                   | 1     |
| W    | Bound Pistol Stock           | 1     |
| T    | Brass Barrel Band            | 1     |
| I    | Tempered Gun Steel           | 1     |
| L    | Leather                      | 1     |

Workshop record: kubejs:flintlocks/pistol

# FK15 Flintlock Rifle

Bring the reinforced barrel and shoulder stock together with the common lock and ramrod. Two further bands, a screw, a tempered piece and a precision mechanism complete the pattern. The rifle is assembled from parts, not from a finished pistol.



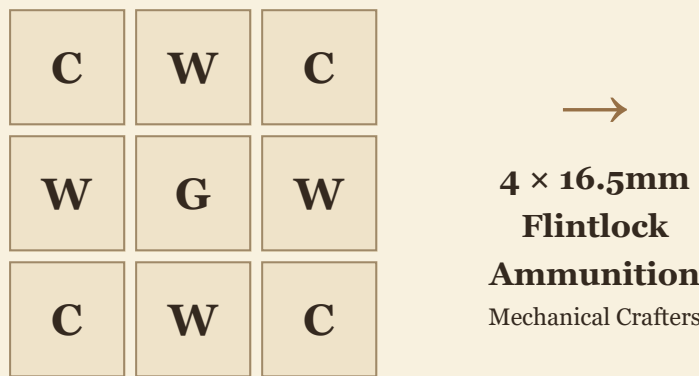
→  
**1 × FK15 Flintlock  
Rifle**  
Crafting table

| MARK | LAY UPON THE BENCH           | COUNT |
|------|------------------------------|-------|
| B    | Brass Barrel Band            | 2     |
| R    | Reinforced Long Barrel       | 1     |
| M    | Complete Flintlock Mechanism | 1     |
| W    | Reinforced Rifle Stock       | 1     |
| S    | Gun Screws                   | 1     |
| T    | Steel Ramrod                 | 1     |
| I    | Tempered Gun Steel           | 1     |
| P    | Precision Mechanism          | 1     |

Workshop record: kubejs:flintlocks/rifle

# 16.5mm Flintlock Ammunition

Set the ammunition work before nine powered Mechanical Crafters. Place large bullet cores in the four corners, wads at the four edges, and one gunpowder charge at the centre. Each completed pattern delivers four game rounds.



| MARK | LAY UPON THE BENCH | COUNT |
|------|--------------------|-------|
| C    | Large Bullet Core  | 4     |
| W    | Wad                | 4     |
| G    | Gunpowder Charge   | 1     |

The display name is 16.5mm; its ammo ID is qkl:16mm. Both guns use it. Each cycle yields four rounds, which stack to sixteen.

Workshop record: kubejs:flintlocks/ammo\_16mm

# Cores, wads & charges

Let the ammunition-maker prepare the three stores separately. Only when all are ready should the nine crafters be filled.

## I. The large cores

A Mechanical Saw cuts one Iron or Zinc Nugget into **10 Large Bullet Cores**. Use Creatified's large cores, not its smaller cores, finished bullets or pellets.

## II. The paper wads

Begin with one Paper. Deploy another Paper upon it three times, press the in-progress piece, then cut it with a Mechanical Saw. This one-pass sequence consumes **4 Paper** and gives **30 Wads**. Carry the same unfinished item through every step.

## III. The prepared charge

Mix **2 Minecraft Gunpowder**, **1 Charcoal** and **250 mB Water** in a Basin to obtain 10 Gunpowder Cakes. No heat is required for this mixing recipe. Smoke each cake into a dry cake, using fuel, then press each dry cake into **4 Gunpowder Charges**.

The smoking step takes 50 ticks, or 2.5 seconds at normal server speed. Five Creatified Gunpowder Pellets can also be crafted shapelessly into one charge.

These names and quantities are game recipes from the installed TaCZ: Creatified mod. The final pattern uses a Gunpowder Charge, not loose vanilla gunpowder.

# The nine-square assembly

Build nine Mechanical Crafters in a 3×3 face. Turn their routes with a wrench so they converge on one output. Supply rotational power and leave an output clear.

Use the ammunition plate: four large cores at the corners, four wads on the edges, and one charge at the centre. Every slot is occupied. Collect four rounds per cycle.

| ROUNDS | CYCLES | CORES | WADS | CHARGES |
|--------|--------|-------|------|---------|
| 4      | 1      | 4     | 4    | 1       |
| 16     | 4      | 16    | 16   | 4       |
| 64     | 16     | 64    | 64   | 16      |

## A worked lot of sixteen rounds

1. Cut two metal nuggets to obtain 20 cores.
2. Use four Paper to make 30 Wads.
3. Mix one batch of ten cakes. Smoke just one cake and press it into four charges.
4. Run the ammunition pattern four times.

The work yields sixteen rounds and leaves four cores, fourteen wads and nine undried cakes. Fuel and machine construction are additional.

## At the close of work

Carry the finished 16.5mm ammunition in your inventory. Hold the gun and use the configured TaCZ Reload key. Both guns hold one round; wait for the reload animation to finish.

# When the work will not answer

| DIFFICULTY             | EXAMINATION                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| No result at the bench | Check every occupied and empty square. Use the exact custom parts, sheets and stripped dark oak logs. |
| Blank will not cook    | Steel and springs need a blast furnace and fuel.                                                      |
| Crafters stand idle    | Check power, stress capacity, routing, all nine ingredients and a clear output.                       |
| Gun will not reload    | Use finished qkl:16mm ammunition in your inventory and check the Reload binding.                      |
| No recipe in EMI       | This is intended, including for admins. Consult the plates.                                           |

## *Here ends the Book of Work.*

Prepared for Rustic Craft II from its installed KubeJS recipe definitions. The plates retain their exact patterns and quantities. This edition uses the workshop's custom item pictures and period-inspired prose.

The original technical reference remains in FLINTLOCK\_CRAFTING\_MANUAL.md. This handbook is a Minecraft game guide, not an authentic historical document.